COMPUTER STORE MARKETPLACE

Database

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About

A good starting database is **MySQL + Spring Boot JPA** with these main tables:

1. Core Database Structure

- users
- roles
- user_roles

- seller_requests
- stores

- categories
- brands

- products
- product_images
- product_specifications

- carts
- cart_items
-
- addresses
- orders
- order_items

- payments
- shipments

- product_reviews
- seller_reviews

- returns
- refunds

- notifications
- platform_settings

Main relationship

```
USER
 │
 ├── USER_ROLES ── ROLE
 │
 ├── SELLER_REQUEST
 │      ↓
 │    STORE
 │      ↓
 │    PRODUCTS
 │      ├── PRODUCT_IMAGES
 │      └── PRODUCT_SPECIFICATIONS
 │
 ├── CART ── CART_ITEMS ── PRODUCT
 │
 ├── ADDRESS
 │
 └── ORDERS
      ├── ORDER_ITEMS ── PRODUCT
      ├── PAYMENT
      ├── SHIPMENT
      ├── RETURN
      └── REFUND

CATEGORY ── PRODUCTS
BRAND ──── PRODUCTS

PRODUCT ── PRODUCT_REVIEWS
STORE ──── SELLER_REVIEWS
```

2. users

Stores all registered users.

Column	Type	Description
id	BIGINT PK	User ID
username	VARCHAR(100)	Username
email	VARCHAR(150) UNIQUE	Email
password	VARCHAR(255)	BCrypt password
first_name	VARCHAR(100)	First name
last_name	VARCHAR(100)	Last name
phone	VARCHAR(30)	Phone
status	VARCHAR(30)	ACTIVE/BLOCKED
created_at	DATETIME	Created date

3. roles

You can start with:

ADMIN, STAFF, CUSTOMER, SELLER

id	name
1	ADMIN
2	STAFF
3	CUSTOMER

4	SELLER

Because one user can potentially have multiple roles, use a separate user_roles table.

user_roles

- user_id → users.id
- role_id → roles.id

Example:

- User A → CUSTOMER
- User A → SELLER

This fits your marketplace because a customer can later become an approved seller.

4. seller_requests

When a customer wants to become a seller:

- USER
- ↓
- seller_requests
- ↓
- ADMIN APPROVAL
- ↓
- SELLER

Column	Type
id	BIGINT PK
user_id	BIGINT PK
business_name	VARCHAR(150)
phone	VARCHAR(30)
address	TEXT
description	TEXT

status	VARCHAR(30)
rejection_reason	TEXT
reviewed_by	BIGINT PK
created_at	DATATIME
reviewed_at	DATATIME

Status:

- PENDING
- APPROVED
- REJECTED
- SUSPENDED

5. stores

After seller approval, the seller creates a store.

Column	Type
id	BIGINT PK
user_id	BIGINT FK → users.id
seller_id	VARCHAR(150)
logo	VARCHAR(255)
description	TEXT
phone	VARCHAR(30)
address	TEXT
status	VARCHAR(30)

created_at	DATETIME

Relationship:

- USER (Seller)
- ▏
- ▏1
- ↓
- STORE
- ▏
- ▏ 1:N
- ↓
- PRODUCTS

6. categories

Admin controls categories.

Column	Type
id	BIGINT PK
name	VARCHAR(100)
description	TEXT
status	VARCHAR(30)

Example:

- Computer
- CPU
- GPU
- RAM
- Motherboard
- SSD
- HDD
- PSU
- Case
- Monitor

- Keyboard
- Mouse
- Accessories

Again, sellers only select an existing brand.

7. brands

Admin controls brands.

Column	Type
id	BIGINT PK
name	VARCHAR(100)
logo	VARCHAR(255)
status	VARCHAR(30)

Example:

- Intel
- AMD
- NVIDIA
- ASUS
- MSI
- Gigabyte
- Corsair
- Kingston
- Samsung
- Western Digital

Again, sellers only select an existing brand.

8. products

This is one of your most important tables.

Column	Type
id	BIGINT PK
store_id	BIGINT PK
category_id	BIGINT PK
brand_id	BIGINT PK
name	VARCHAR(150)
description	TEXT
price	DECIMAL(12,2)
stock	INT
condition	VARCHAR(30)
warranty	VARCHAR(100)
status	VARCHAR(30)
approval_reason	TEXT
created_at	DATETIME
updated_at	DATETIME

Example:

- Store: ABC Computer
- Category: CPU
- Brand: AMD

- Name: Ryzen 7 7800X3D
- Price: $399
- Stock: 10

9. product_images

Don't put 8 image columns directly inside products.

❌ Don't do:

- image1
- image2
- image3
- ...
- image8

Instead:

- products
- ↓
- product_images

Column	Type
id	BIGINT PK
product_id	BIGINT PK
image_url	VARCHAR(500)
sort_order	INT
created_at	DATETIME

Then one product can have:

- Product 1
- ├── image 1
- ├── image 2
- ├── image 3
- ├── image 4
- └── ...
- └── image 8

The **8-image limit should be a platform setting/business rule**, not eight columns.

10. Product Specifications

This part is important for your project because different computer categories have different specifications.

You have two common approaches.

Option A — Flexible specification tables

- specification_definitions
- product_specifications

specification_definitions

id	category_id	name
1	CPU	Socket
2	CPU	Cores
3	CPU	Threads
4	CPU	Base Clock
5	CPU	Boost Clock
6	RAM	Capacity
7	RAM	DDR Type
8	RAM	Speed

product_specifications

id	product_id	specification_id	value
1	100	1	AM5
2	100	2	8
3	100	3	18
4	100	4	4.2 GHz

This allows your admin to define specifications per category.

For example:

- CPU
- ├── Socket
- ├── Cores
- ├── Threads
- ├── Base Clock
- └── Boost Clock
- RAM
- ├── Capacity
- ├── DDR Type
- ├── Speed
- └── Modules

For your marketplace, this is much more flexible than adding columns such as cpu_socket, ram_speed, etc. to the product table.

11. addresses

Customers need delivery addresses.

Column	Type
id	BIGINT PK
user_id	BIGINT FK

recipient_name	VARCHAR(150)
phone	VARCHAR(30)
province	VARCHAR(100)
district	VARCHAR(100)
commune	VARCHAR(100)
address_detail	TEXT
is_default	BOOLEAN

One user can have multiple addresses.

- USER
- ├── Address 1
- ├── Address 2
- └── Address 3

12. carts

Column	Type
id	BIGINT PK
user_id	BIGINT FK
created_at	DATETIME
updated_at	DATETIME

cart_items

Column	Type
id	BIGINT PK
cart_id	BIGINT FK
product_id	BIGINT FK
quantity	INT

Relationship:

- USER
- ↓
- CART
- ↓
- CART_ITEMS
- ↓
- PRODUCT

13. orders

Column	Type
id	BIGINT PK
user_id	BIGINT FK
address_id	BIGINT FK
order_number	VARCHAR(50) UNIQUE
subtotal	DECIMAL(12,2)
shipping_fee	DECIMAL(12,2)

total_amount	DECIMAL(12,2)
status	VARCHAR(30)
created_at	DATETIME

Order status:

- PLACED
- PAYMENT_CONFIRMED
- PROCESSING
- SHIPPED
- DELIVERED
- COMPLETED
- CANCELLED

14. order_items

This is important for a marketplace.

One order can contain products from different sellers.

Column	Type
id	BIGINT PK
order_id	BIGINT FK
product_id	BIGINT FK
store_id	BIGINT FK
product_name	VARCHAR(150)
price	DECIMAL(12,2)
quantity	INT

subtotal	DECIMAL(12,2)

For example:

- Order #1001
- ├── RTX 5070
- │ Seller A
- │
- ├── RAM 32GB
- │ Seller B
- │
- └── SSD 1TB Seller C

This is one of the differences between a normal computer store and a **marketplace**.

15. payments

Column	Type
id	BIGINT PK
order_id	BIGINT FK
payment_method	VARCHAR(50)
transaction_id	VARCHAR(255)
amount	DECIMAL(12,2)
status	VARCHAR(30)
paid_at	DATETIME

Example:

- PENDING
- PAID
- FAILED

- REFUNDED

16. shipments

Column	Type
id	BIGINT PK
order_id	BIGINT FK
shipping_method	VARCHAR(100)
tracking_number	VARCHAR(100)
status	VARCHAR(30)
shipped_at	DATETIME
delivered_at	DATETIME

17. product_reviews

Column	Type
id	BIGINT PK
product_id	BIGINT FK
user_id	BIGINT FK
order_item_id	BIGINT FK
rating	INT
comment	TEXT

created_at	DATETIME

Rating:

- 1 - 5

You should connect the review to order_item so you can enforce:

- Only customers who actually purchased the product can review it.

18. seller_reviews

For rating the store/seller separately.

Column	Type
id	BIGINT PK
store_id	BIGINT FK
user_id	BIGINT FK
order_id	BIGINT FK
rating	INT
comment	TEXT
created_at	DATETIME

So you have:

- Product Review
- ↓
- Product Seller Review
- ↓
- Store/Seller

19. returns

Column	Type
id	BIGINT PK
order_item_id	BIGINT FK
user_id	BIGINT FK
reason	TEXT
status	VARCHAR(30)
requested_at	DATETIME
reviewed_at	DATETIME

Example:

- REQUESTED
- APPROVED
- REJECTED
- RECEIVED
- COMPLETED

20. refunds

Column	Type
id	BIGINT PK
order_id	BIGINT FK

return_id	BIGINT FK
amount	DECIMAL(12,2)
reason	TEXT
status	VARCHAR(30)
refunded_at	DATETIME

21. platform_settings

This is where your admin-controlled rules can be stored.

id	setting_key	setting_value
1	max_product_images	8
2	max_image_size_mb	5
3	max_product_name_length	150
4	max_description_length	5000
5	seller_approval_required	true
6	product_approval_required	true

This is better than hard-coding every rule.

22. notifications

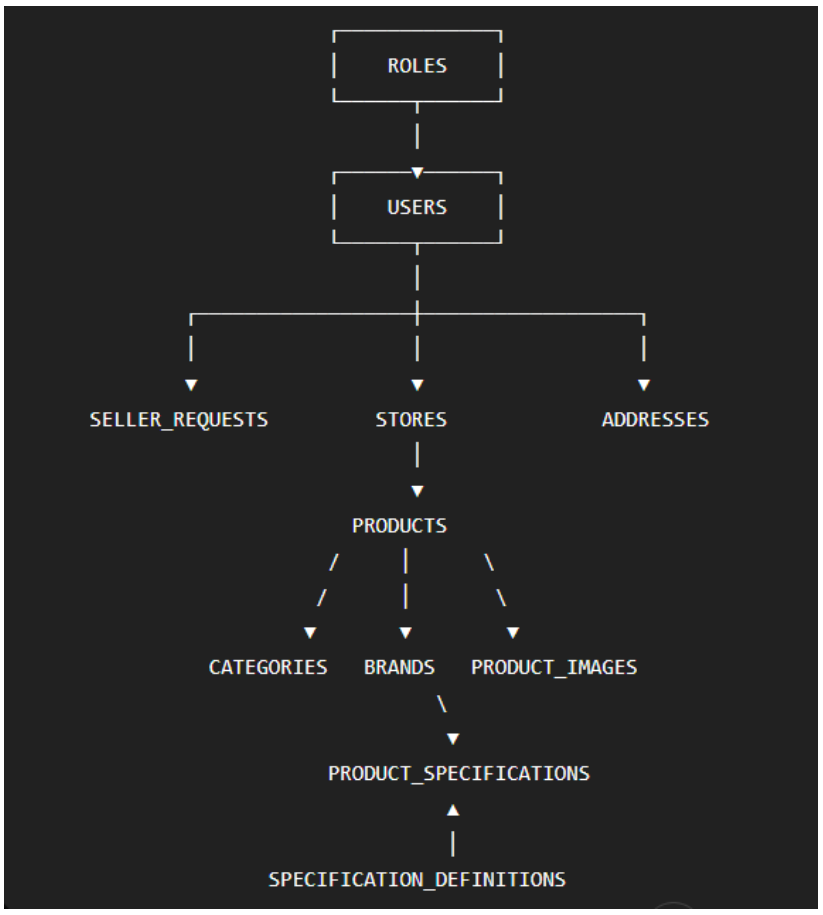
Useful for your approval workflow.

Column	Type
id	BIGINT PK
user_id	BIGINT PK
title	VARCHAR(200)
message	TEXT
type	VARCHAR(50)
is_read	BOOLEAN
created_at	DATETIME

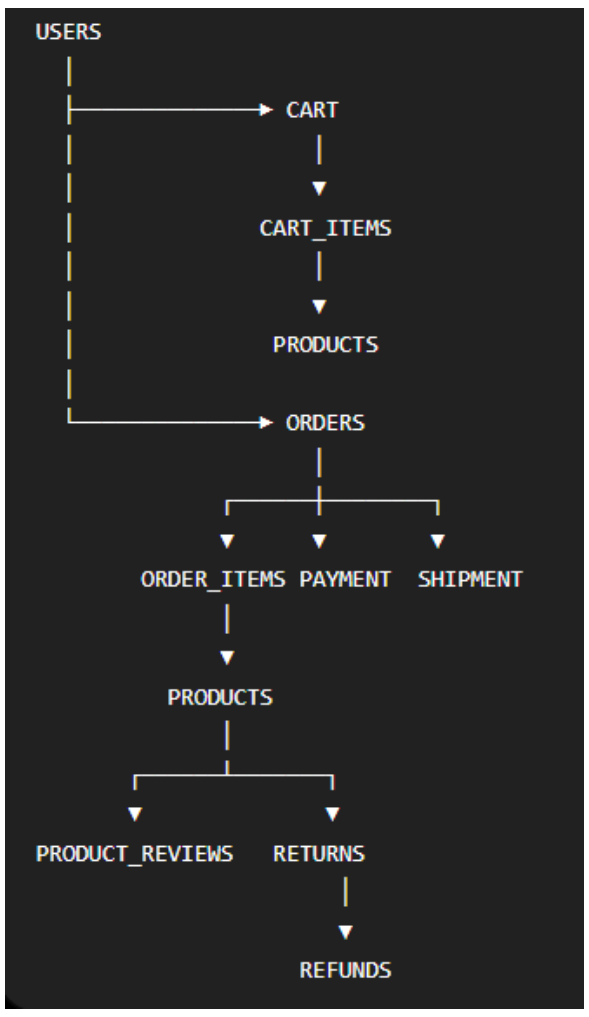
For example:

- "Your seller request has been approved."
- "Your product was rejected."
- "Your order has been shipped."

Complete Database ER-style Flow



Then the customer side:



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Topic : Computer Store Marketplace

Subject : Java III